



SCI_LAUNCH_A2_W01 · teacher note

Objective (verbatim): Describe mitosis & the cell cycle (animation).

Scheme of work: LAUNCH scheme of work, Autumn 2 week 1.

Pupil words: I am learning to order the cell cycle and explain why a cell makes two matching cells.

Reading gate: LAUNCH 11+; ≈14.7 years (rough visible-body ARI grade + 6; 895 pupil-visible words, excluding data-mbm-guide). This is a rough text-screening proxy, not a pupil's measured reading age. Technical terms and short task directions make different formulas disagree.

Specification: Shared Pearson Edexcel GCSE Biology 1BI0 Foundation / Combined Science 1SC0 Foundation Topic 2 points 1.1 and 2.1–2.3. The large-stage diagram plus full and faded phase sequence name interphase, prophase, metaphase, anaphase, telophase and cytokinesis from 2.1. Growth/repair and matching body cells address 2.2–2.3. This one lesson does not teach asexual reproduction, full molecular detail or the term diploid. The drawn symbols are simplified; use the official Biology specification (<https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf>), Combined Science specification (<https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>) and Pearson mark scheme (<https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Exam-materials/1bi0-1h-rms-20240822.pdf>) for more detailed phase descriptions.

Gatsby 4: A laboratory technician labels a cell sample and records what its microscope image shows.

Sequence and explicit instruction

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets sit inside We do, Independent and Review; arrival and resets make about eight minutes for settle/transition. No added time.

Success criteria at Starter, checked at Exit: I can order interphase, four mitosis phases and cytokinesis.; I can show that the two new body cells have matching chromosome sets.; I can explain how mitosis helps growth or repair.

Retrieval from the actual SoW sequence:

“Last term” means the preceding Autumn 1 teaching block in this SoW. If it means a prior academic term before teaching, that source is unavailable here and this row needs a staff check rather than a guessed prior-year lesson.

When	Question	Answer	Source cell
last lesson	Which process uses energy to move a substance from low to high concentration?	Active transport.	LAUNCH scheme of work, Autumn 1 week 7 (Aut1·W7 review); LAUNCH scheme of work, Autumn 1 week 6 (Aut1·W6 active transport)
2-4 weeks	In osmosis, which substance moves through a partially permeable membrane?	Water.	LAUNCH scheme of work, Autumn 1 week 5 (Aut1·W5 osmosis)
last term	In a eukaryotic animal cell, which structure holds the chromosomes?	The nucleus.	LAUNCH scheme of work, Autumn 1 week 1 (Aut1·W1 eukaryotic and prokaryotic cells)

Vocabulary (same words, definitions, examples as lesson and organiser):



Term	Tier	Definition	Example sentence
cell cycle	3	the series of steps in which a cell grows and divides	The cell cycle includes interphase, mitosis and cytokinesis.
interphase	3	the stage in which the cell grows and copies its DNA	The cell copies its DNA in interphase.
mitosis	3	the stage in which copied chromosomes separate into two nuclei	Mitosis separates the chromosome sets.
chromosome	3	a structure of DNA that carries genes	Each new nucleus receives a matching set of chromosomes.
cytokinesis	3	the division of the cell after the nuclei form	Cytokinesis splits one cell into two.
prophase	3	the mitosis phase when chromosomes condense	In prophase, chromosomes become easier to see.
metaphase	3	the mitosis phase when chromosomes line up in the middle	In metaphase, chromosomes form a line.
anaphase	3	the mitosis phase when the two copies of each chromosome are pulled to opposite ends	In anaphase, the two copies of each chromosome are pulled to opposite ends.
telophase	3	the mitosis phase when two new nuclei form	In telophase, each set gains a nucleus.

Faded worked example: I do full model: Interphase: copy DNA. Mitosis: prophase (condense), metaphase (line up), anaphase (the two copies of each chromosome are pulled to opposite ends), telophase (two nuclei). Cytokinesis: split the cell. We do part-done: Interphase: copy DNA. Mitosis: prophase (condense), metaphase (line up), , telophase (two nuclei). Cytokinesis: split the cell. The removed step is Anaphase: the two copies of each chromosome are pulled to opposite ends. Independent task uses the model without a filled answer. Fade next time: Next lesson remove the metaphase cue as well; ask pupils to rebuild the sequence from the model before checking.

Check for understanding after I do: "Show me: is DNA copied before, during or after mitosis?" Expected: Before mitosis, in interphase. Pupils show a whiteboard answer or point to the first panel; re-model the paper strips if the response is insecure.

Challenge and stretch

We do support: Use the full model beside the faded one. Point to the two nuclei before choosing the missing step.

We do stretch: Explain why swapping mitosis and DNA copying would break the model.

Independent F: Order the three cards and show the same chromosome colours in each new cell. Stem: First ...; then ...; last ...

Independent M: Explain why each new cell has a matching chromosome set. Stem: Each new cell matches because ...

Independent H: Predict what would happen to the sets if copied chromosomes did not separate evenly. Stem: If the copies did not separate evenly, then ... because ...

Go deeper: How do copy, separate and split work together when new cells are needed for repair?



Named early-finisher stretch — Fault finder: Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. It compares two distinct faults and proposes repairs, beyond the H prediction of a single unequal separation.

Hinge and misconceptions

We do 2 question: A new body-cell model starts with six chromosomes. After DNA copying, mitosis and cell division, what should each daughter cell contain?

Correct C: C. The same six-chromosome set as the original body cell.

Three wrong-answer diagnoses: A — Confuses body-cell mitosis with chromosome halving in meiosis. B — Counts the copied chromosomes before separation as the number retained by each daughter cell. D — Assumes copied chromosomes stay in the parent cell instead of separating into two cells.

Decision: About 3 in 4 or more right → move on; otherwise re-model copying and splitting with two blue and two orange paper strips, then ask the parallel hinge. No automatic grade or saved response.

Parallel hinge: A different body-cell model starts with eight chromosomes. After ordinary mitosis and cell division, how many are in each daughter? Answer: eight, with a matching set.

Two anticipated misconceptions:

Misconception	Spot it	Correct it
Mitosis halves the chromosome number to make gametes.	Pupil chooses A, the halved three-chromosome set.	Contrast mitosis with meiosis: for this body-cell model, copy and separate so both cells keep the same chromosome set.
Mitosis alone is the whole cell split.	Pupil chooses D, leaving copies in one cell.	Point to the final cytokinesis arrow; move the two paper circles apart to represent two cells.

Two on-screen interactions: three-step sequence with keyboard-native selects, an explicit check and polite live feedback; four-option all-pupil hinge, answered simultaneously by whiteboard, symbol or point and discussed, without an automatic grade. JS holds only transient form state; reset clears it.

Planned adaptations

Need	Concrete adaptation	EEF five-a-day strategy
Cognition & Learning	Use three large cards with a four-phase mini-strip inside mitosis; remove the anaphase cue, then fade metaphase next lesson.	scaffolding
Communication & Interaction	Give two quiet partner turns and the same speaking frame before any writing; accept point, sign or AAC for the explanation.	flexible grouping
SEMH	Check in before the top rung and hinge; offer an agreed pause and a one-step return without public correction.	cognitive/metacognitive strategies
Sensory/Physical	Offer enlarged, high-contrast, easy-grip chromosome strips and a seated or standing choice; accept speech-to-text.	technology



Need	Concrete adaptation	EEF five-a-day strategy
EAL (language access, not SEN)	Pre-teach the science terms with diagram and example sentences; allow planning in any language and the same M/H scientific task with a speaking frame.	explicit instruction

EAL support remains at M and H: a diagram and speaking frames reduce language load without reducing scientific challenge. Talk before write. The “My language” organiser column is blank for the pupil; no machine translation is supplied. Offer text-to-speech, immersive reader and speech-to-text where available.

TA Focus, prompting and independent stage

At lesson start: choose the teacher or TA to work with supported pupils at Independent; the other holds the group. Swap roles across the week. The adult with the group can still ask the cold-call bank; no public demand on an individual pupil to disclose a need.

TA line at I do: “Watch the DNA copy. Point to where the two matching sets go.”

TA line at We do: “You have the first and last steps. What might happen between them?”

TA line at Independent: “Show me the last step you know. What does the model tell you next?”

Prompt ladder, least first; no physical prompting:

1. Wait: count to five and watch for an independent move.
2. Point or gesture to the last complete card or model step.
3. Verbal cue: “What happened to the copied chromosomes?”
4. Model one move with one colour; pupil repeats with the other colour.
5. Finish one step together; then ask the pupil to try a fresh step alone.

Cold-call bank (adult use, two per tier):

Tier	Question	Answer
F	Which stage copies DNA?	Interphase.
F	Which step splits the whole cell?	Cytokinesis.
M	Why do the daughter cells have matching sets?	DNA was copied and the matching chromosome sets separated into two nuclei.
M	How does mitosis help growth?	It produces more body cells with matching chromosome sets.
H	What happens to the two copies of each chromosome during anaphase?	The two copies of each chromosome are pulled to opposite ends.
H	Why is the order copy, separate, split important?	The copies must exist before separation, and the cell divides after each nucleus has a set.

Response modes at Independent: point, sign, speak, draw, type, write, scribe or staff-captured photo. Record the actual scientific step and prompt level, not the access aid itself. Handoff at Exit: tell the other adult the prompt level reached, what was independent, the feedback given and the next step; the teacher uses the exit answer to choose the following model, question or prompt.

Feedback, regulation and evidence



Code set: VF · WS · I · NS(+) · E · R · // · ?

When: VF at model/Review; WS when a scaffold is used; I at unsupported Independent/Exit; NS(+) is a precise next step at Exit; // is a real pupil self-edit at Review; ? is read-back or communication of that change; R is the next-lesson response to NS(+) heard by an adult; E only where a genuine observed outcome matches a current plan.

What I record and where: note the hinge distribution and named misconception on the staff quick-record grid; use actual codes in feedback records. E: evidence captured on the school's evidence app. Do not save scores or pupil data in HTML.

Pathways: high support, VF/WS often and first simple next steps. Build: WS/I mix and explains feedback. Grow: mostly independent, self-edits and responds. Launch: independent self-assessment with evidence.

Regulation: Check in before hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Point to the last completed step at re-entry and offer a small structured choice. Correct privately.

QR poster: LAUNCH_SCI_A2_W01, staff device or projector only. No pupil phone or embedded external load.

Lundy Exit: pupil says, points, signs, draws or writes; the teacher hears the answer and chooses the next example or prompt. Four pupil-voice answers are available: learning goal (header), actual VF (Review), NS(+) (Exit), success criteria (Starter/Exit).

Source check and limitations

Facts are enumerated with official specification points and URLs in LESSON_SOURCE.json. Distractor sentences are deliberately false and corrected in the hinge diagnoses. The diagram is an original simplified overview of the large stages; the full/faded strip adds the four named mitosis phases. The old “long package of DNA” definition has been tightened to a DNA structure carrying genes.

Practical: the paper-strip activity is a model; the lesson library lessons SCI_L_W9L2 and SCI_L_W9L3 animate every mitosis phase with a reduced-motion option in the same week.

Current hinge options

Option	Correct?	Diagnosis
A. Half the original set, with just three chromosomes each.	no	Confuses body-cell mitosis with chromosome halving in meiosis.
B. Double the original set, with twelve chromosomes in each.	no	Counts the copied chromosomes before separation as the number retained by each daughter cell.
C. The same six-chromosome set as the original body cell.	yes	Uses the new case accurately.
D. No chromosomes, because the copies remain in one cell.	no	Assumes copied chromosomes stay in the parent cell instead of separating into two cells.

Prepared model handling · practical safety

Kit: Large prepared paper chromosome strips, two cell outlines, tray

Method: Place the prepared strips into the model cell outlines and check the matching sets.

Pupil Risk: Loose paper strips could fall or be mouthed.

Pupil Safe: Use large prepared strips at a desk. Keep them out of mouths and collect dropped pieces.



Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W01

SoW reference: LAUNCH scheme of work, Autumn 2 week 1

Exact SoW outcome: Describe mitosis & the cell cycle (animation).

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W01. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Point to the first and last picture in the cell strip. What changed? Staff: Offer the same strip as large print or tactile paper shapes; accept pointing before speaking.

Starter (3 min): Recall three things from earlier lessons. Show one answer at a time. Staff: Ask the three retrieval questions in order, reveal answers after all pupils respond, then point to the three success criteria.

I Do (5 min): Watch the full model: copy the DNA, separate the matching chromosomes, then split the cell. Staff: Show the paper-strip model; pause for the show-me check before the next stage.

We Do (7 min): Complete the missing step in the part-done model. Then place the paper strips in both new cells. Staff: Run paired talk before the card or select task; offer the support route without giving the missing step.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Use simultaneous whiteboards, point or symbol cards. Apply the decision rule and parallel hinge in the note.

Independent (8 min): Use the model to explain how cell division helps replace damaged body cells. Staff: One adult supports pupils; the other holds the group, swapping roles across the week. Record actual independence and prompt level.

Review (4 min): Tell your partner what you changed after feedback. Mark a real self-edit // and read it back when asked ?. Staff: Give one specific piece of VF, allow a true self-edit and record only feedback that occurred.

Exit (3 min): Show how you know: order the three stages and explain why the new cells match. Staff: Hear every response through speech, sign, point, drawing or writing; use it to change the next example or prompt.

Planned adaptation — cognition learning: Use three large cards with a four-phase mini-strip inside mitosis; remove the anaphase cue, then fade metaphase next lesson.

Planned adaptation — communication interaction: Give two quiet partner turns and the same speaking frame before any writing; accept point, sign or AAC for the explanation.

Planned adaptation — semh: Check in before the top rung and hinge; offer an agreed pause and a one-step return without public correction.

Planned adaptation — sensory physical: Offer enlarged, high-contrast, easy-grip chromosome strips and a seated or standing choice; accept speech-to-text.



Planned adaptation — eal: Pre-teach the science terms with diagram and example sentences; allow planning in any language and the same M/H scientific task with a speaking frame.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Use the model to explain how cell division helps replace damaged body cells. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W02. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W02: Explain growth & stem cells.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Anaphase: the two copies of each chromosome are pulled to opposite ends. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A new body-cell model starts with six chromosomes. After DNA copying, mitosis and cell division, what should each daughter cell contain?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Point to the first and last picture in the cell strip. What changed? Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Use the model to explain how cell division helps replace damaged body cells. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use cell cycle, interphase, mitosis, chromosome, cytokinesis, prophase, metaphase, anaphase, telophase with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.

Source detail: Pearson Edexcel GCSE Biology 1BI0 points 1.1 and 2.3 describe the nucleus and identical chromosome sets: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf>. Osmosis is the net movement of water from a dilute to a more



concentrated solution through a partially permeable membrane: <https://www.aqa.org.uk/subjects/biology/gcse/biology-8461/specification/subject-content/cell-biology>.